



Department of Information Technology

Academic Year 2025 – 2026(Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT

Course Code & Title: CS3391 – Object Oriented Programming

Name of the Faculty member (s): Dr. K. Palraj, ASP / IT

Innovative Practice Description

- **Unit / Topic:** Unit 4 / Strings: Basic String class, methods
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO14.
- **Activity Chosen:** Cross Word Puzzle

- **Justification:**

A crossword puzzle based on Java String class and methods is an effective learning tool because it actively engages students in recalling key concepts such as method names, their syntax, and their purpose, thereby reinforcing fundamental programming knowledge in an interactive way. It promotes critical thinking, vocabulary familiarity, and accurate spelling of method names like substring, charAt, equals, and trim, which are essential for writing error-free code. By transforming learning into a structured puzzle activity, it increases student motivation, reduces learning monotony, and serves as a low-stress formative assessment tool to evaluate understanding without direct coding. This activity can also be used to initiate discussions, identify misconceptions, and support peer learning, making it a valuable addition to classroom teaching strategies for computer programming.

- **Time Allotted for the Activity:** 10 Minutes

- **Details of the Implementation:**

- The activity was conducted in a classroom for a duration of 10 minutes. The students were given 14 questions as a crossword puzzle in the topic of string and string functions.
- They have to think and write the answers in the given puzzle.
- The activity was conducted during the last 10 minutes of the class after that the answers were corrected. Sample answers were shown in Figure 1, Figure 2 & Figure 3.
- The answers for the wrong questions were discussed at the end of the class to make them to know the answer of all the questions.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	3	3	3	3	3	3	3	3	3	3	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
	3	3	3	3	3	3	3	3	3	3	3	2
Justification for correlation	Students will be able to apply core Java programming concepts such as I/O, strings, collections, and generics to develop efficient applications	Students will be able to analyze user requirements and select suitable Java features to read, process, and store data	Students will be able to develop structured Java programs using collections and generics to meet given specifications	Students will be able to experiment with different Java APIs to evaluate performance, memory use, and code reusability	Students use IDEs, debugger, and compilers (e.g., Eclipse, IntelliJ) to write and test object-oriented programs	Students will be able to enhance coding skills individually and through teamwork in building modular Java applications	Students will be able to explain their programs using comments, diagrams, and presentations to share their ideas clearly	Students will be able to understand the importance of time and resource management when developing Java applications as part of a project.	Students could be able to encourage continuous exploration of new Java libraries, updates, and best practices in programming	Students will be able to use collections and generics to store, process, and manipulate large data efficiently	Students will be able to build the programming base required in areas like Android development, server-side programming, and data processing frameworks	Students will be able to develop maintainable, reusable, and clean code using Java's core libraries and OOP principles

- **Images / Screenshot of the practice:**

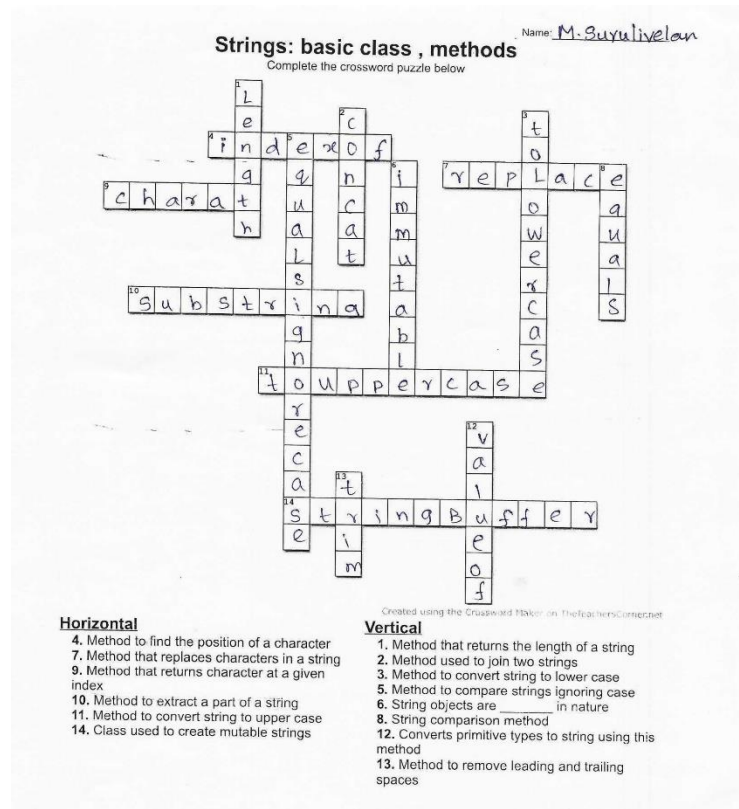


Figure 1. Crossword problem solved by M. Surulivelan

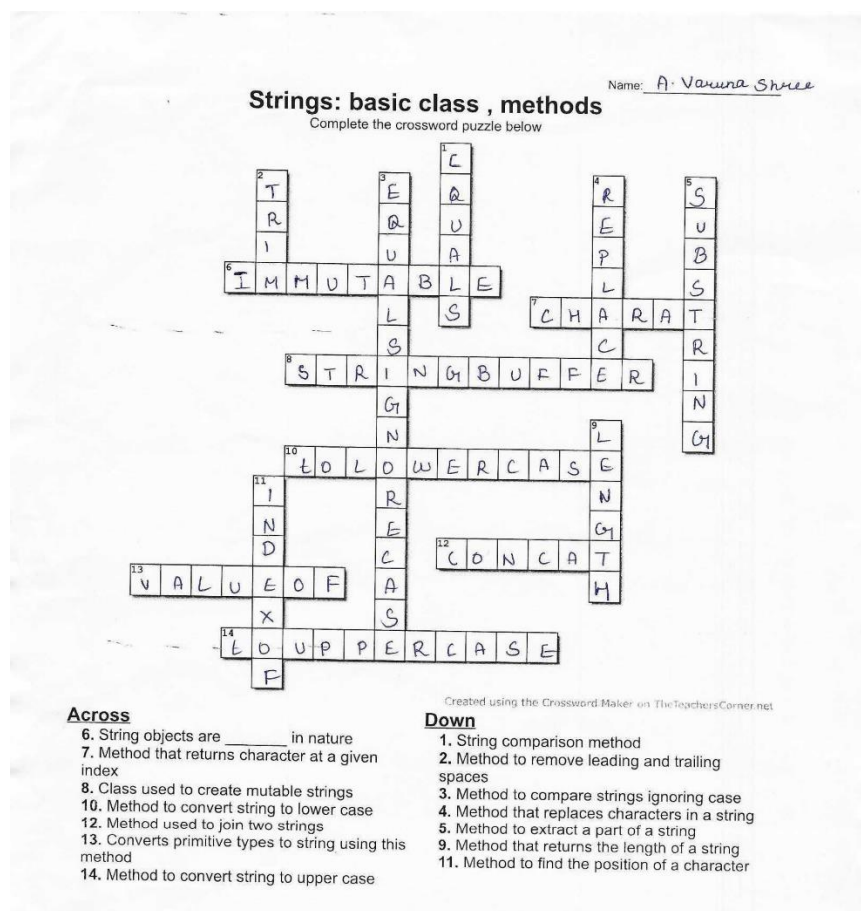


Figure 2. Crossword problem solved by A. Varuna Shree

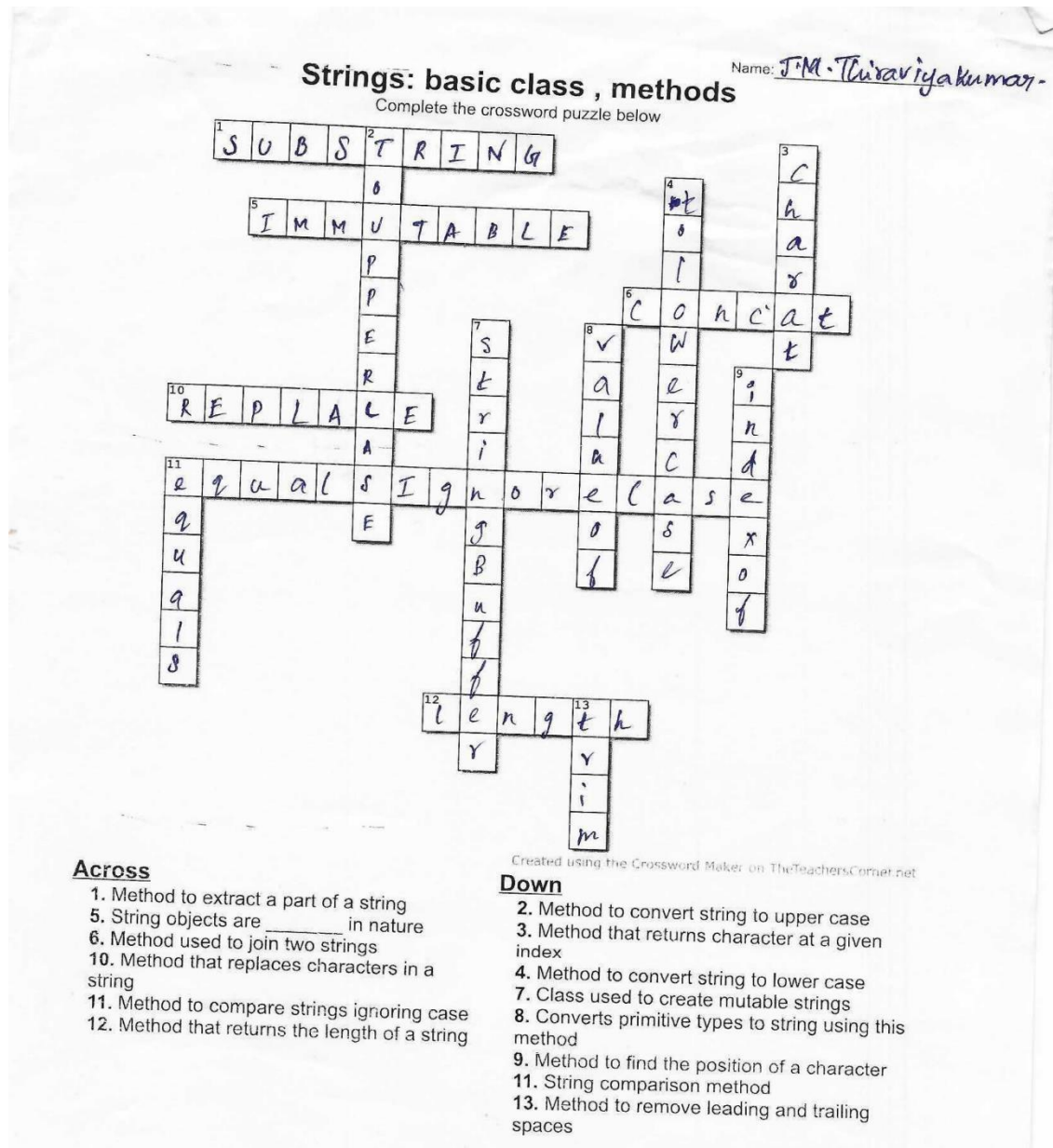


Figure 3. Crossword problem solved by J.M. Thiraviyakumar

• **Reflective Critique:**

• **Feedback of practice from students and other stakeholders:**

- ✓ Students reported that the crossword puzzle made learning String functions more interesting and enjoyable compared to traditional lectures.
- ✓ Some students mentioned that the crossword helped reduce exam fear and encouraged active participation.
- ✓ It helped identify slow learners and give them support in a friendly and fun learning environment.

- ***Benefit of the practice:***

- Students developed a better understanding of String methods and their usage (e.g., substring(), equals(), compareTo()) through repeated recall and contextual clues.
- The puzzle format utilized active recall, helping students remember key functions and syntax for a longer duration.
- Students used logical thinking and analytical skills to deduce clues and fill answers, improving their reasoning abilities.
- The game-based method reduced fear of programming and made technical topics like strings easy and enjoyable

- ❖ ***Challenges faced in implementation:***

- Some students were hesitant to participate initially, considering puzzles as non-academic or unrelated to programming.
- Some students struggled with similar string functions (e.g., equals() vs ==, or substring() indexes), which caused confusion during the activity.
- Maintaining discipline and keeping students focused during an interactive activity required strong classroom control.

References:

- https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/GSS_CS3251_Crossword_puzzle.pdf
- https://wordmint.com/public_puzzles/1248202
- https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/AAL_CS3551_Crossword_Puzzle.pdf

Signature of Faculty Member

HOD